Structured finite-window response sensitivity in correlated Hubble dipoles

Anonymous research draft
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Four retained redshift-window dipoles carry a 12-component correlated discrepancy after fitting a bulk response and constant directional rate. We ask how much of this discrepancy can be moved by three prespecified finite-window response classes. Under the retained Gaussian working covariance, the baseline quadratic discrepancy is 23.3967, above the fixed six-degree 95% reference 12.5916. A depth-slope response leaves an irreducible 15.3160 and cannot reach that reference at any coefficient budget. An FP-composition proxy reaches it at a joint dimensionless coefficient radius 5.7509; a six-column combination reaches it at 1.5897. These are exact finite-dimensional sensitivity budgets for declared designs, not estimates of calibration error. We derive the image ellipsoid, its support and attainment, and the minimum-budget crossing including an irreducible-floor equality case.

I. RETAINED CARRIER AND PHYSICAL QUESTION

The saved four-window analysis represents each angular-rate estimator by a monopole, three Cartesian dipole coordinates, and five signed symmetric trace-free coordinates in ICRS J2000. The centers are $z_w = (0.025, 0.040, 0.060, 0.080)$ with saved width $h = 0.012$. Its full 36-by-36 joint covariance connects axes and windows, including shared groups. We extract the marginal 12-by-12 dipole covariance C and window-major signed dipoles y; no conditioning on measured monopole or STF coordinates is performed. The complete physical coefficient carrier and source covariance are retained in the accompanying `source_carrier.json`, alongside the extracted y, C and response matrices [1, 2]. This extraction does not replace those tensors by a scalar.

The retained estimator is a finite weighted angular-rate fit, $cz_i = d_{\mathrm{DM},i}[H_w + \mathbf{D}_w \cdot \mathbf{n}_i + \mathbf{n}_i^{\mathsf{T}} Q_w \mathbf{n}_i]$. Its coefficients are in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$; they are not themselves physical expansion coefficients. The source contains 35,970 groups, assumes a diagonal row-scatter working law, and leaves common calibration, peculiar-velocity and selection covariance unmeasured. The experiment here consumes only the saved statistics. It does not refit a catalogue or infer an observer-frame or cosmographic transfer.

Let R stack the saved nine-by-three bulk-velocity dipole response blocks, with units Mpc^{-1}, and let $G = \mathbf{1}_4 \otimes I_3$. The six-column baseline $X = [R, G]$ has full rank. Its model is

$$y = Rb + Gd + D\eta + e, \qquad e \sim N(0, C) \quad \text{(retained working law)}. \tag{1}$$

Here b is in $\mathrm{km\,s^{-1}}$; d, y, e are in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. The coefficient η is dimensionless under our chosen D. The working Gaussian tail at the baseline is 6.7393×10^{-4}, a conditional diagnostic under the saved covariance, not survey-null coverage.

II. THREE FIXED RESPONSE MECHANISMS

We specify a rate unit $u_0 = 1\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, a depth coordinate $t_w = (z_w - 0.04)/h = (-1.25, 0, 5/3, 10/3)$, and the retained FP distance-weight fractions

$$p = (0.30371471, 0.53491724, 0.80241152, 0.90188987), \qquad f_w = p_w - p_0. \tag{2}$$

For window w and Cartesian component i, each response has one coefficient per axis a:

$$(D_z)_{(w,i),a} = u_0 t_w \delta_{ia}, \quad (D_{\mathrm{FP}})_{(w,i),a} = u_0 f_w \delta_{ia}, \quad D_{\mathrm{joint}} = [D_z, D_{\mathrm{FP}}]. \tag{3}$$

D_z is a linear finite-center directional-rate slope per saved width. D_{FP} is a dipolar offset modulated by FP composition; subtraction of p_0 changes only a constant-rate column already fitted by G. The composition fractions do not give a signed distance-modulus calibration derivative. The joint design has six coefficients in depth-first, then FP, Cartesian order. All three are declared before seeing their residual reduction, and none is preferred as a physical error model on that basis. The common coefficient metric for each dimension is $M = I_p$: $\|\eta\|_2 \le \rho$. This is one joint sensitivity budget, not independent axis allowances or an empirical prior.

III. PROJECTION, ELLIPSOID AND MINIMUM BUDGET

Write $C = LL^{\mathsf{T}}$ without jitter, $W = L^{-1}$, $A = WX$, and let Q_A have orthonormal columns spanning A. The orthogonal residual projector is $P = I - Q_A Q_A^{\mathsf{T}}$. With $r = PWy$, $B = PWDM^{-1/2}$, and $\xi = M^{1/2}\eta$, profiling the six signal coefficients reduces Eq. (1) to

$$T_{\min}(\rho) = \min_{\|\xi\| \le \rho} \|r - B\xi\|^2, \quad T_0 = \|r\|^2 = 23.3967205227. \tag{4}$$

The covariance is the complete dipole marginal; using only within-window blocks would produce $T_0 = 8.92571218$ and change the question. We retain the original six-degree reference $\tau = \chi^2_{6,0.95} = 12.5915872437$ after minimizing over ξ. Replacing its degree count by $12 - \operatorname{rank}[X, D]$ would create a different working-law test.

The image $\mathcal{E}_\rho = \{B\xi : \|\xi\| \le \rho\}$ is generally a degenerate ellipsoid in the 12D whitened space. Cauchy–Schwarz gives its exact support in direction u:

$$h_{\mathcal{E}_\rho}(u) = \rho\|B^{\mathsf{T}}u\|, \quad \xi_* = \rho\frac{B^{\mathsf{T}}u}{\|B^{\mathsf{T}}u\|} \quad (B^{\mathsf{T}}u \neq 0). \tag{5}$$

When $B^{\mathsf{T}}u = 0$, support is zero and $\xi = 0$ attains it. A proposed correction q is attainable exactly when $q \in \operatorname{col} B$ and $\|B^{+}q\| \le \rho$: the pseudoinverse is the minimum-norm preimage, while adding a kernel vector cannot lower its norm. An unrestricted 12D ball omits the column-space condition.

For the resolved thin singular-value decomposition $B = U\operatorname{diag}(\sigma_i)V^{\mathsf{T}}$, define $\alpha = U^{\mathsf{T}}r$ and $r_\perp = r - U\alpha$. The signed $r_\perp$ and all singular vectors are exported with the data. Orthogonality gives the irreducible discrepancy $T_\perp = \|r_\perp\|^2$ and saturation radius $\rho_{\mathrm{sat}} = \|\operatorname{diag}(\sigma_i^{-1})\alpha\|$. For $0 < \rho < \rho_{\mathrm{sat}}$, stationarity of the convex Lagrangian $\|r - B\xi\|^2 + \lambda(\|\xi\|^2 - \rho^2)$ yields

$$\xi_\lambda = V\operatorname{diag}\left(\frac{\sigma_i}{\sigma_i^2 + \lambda}\right)\alpha, \qquad \rho^2 = \sum_i \frac{\sigma_i^2\alpha_i^2}{(\sigma_i^2 + \lambda)^2}, \tag{6}$$

$$T_{\min}(\rho) = T_\perp + \sum_i \frac{\lambda^2\alpha_i^2}{(\sigma_i^2 + \lambda)^2}. \tag{7}$$

The multiplier is positive and unique on the active interval. At zero budget $T_{\min} = T_0$; at or above saturation the minimum-norm solution $B^{+}r$ attains $T_\perp$. If $B = 0$, the discrepancy never moves. Hence the least coefficient budget for $T_{\min} \le \tau$ is zero when $T_0 \le \tau$, unattainable when $T_\perp > \tau$, and ρ_{sat} when $T_\perp = \tau < T_0$. The last case is finite attainment, including equality. In the strict crossing case $T_\perp < \tau < T_0$, a unique positive multiplier solves Eq. (7) at τ, with ρ obtained from Eq. (6).

A second budget counts the image $q = \|B\xi\|$, in whitened discrepancy units. Its exact least value for the same threshold inside $\operatorname{col} B$ is

$$q_{\min} = \sqrt{T_0 - T_\perp} - \sqrt{\tau - T_\perp}, \qquad T_\perp \le \tau < T_0. \tag{8}$$

This follows by moving radially toward $U\alpha$ within the image subspace. It differs from the image norm $q_{95} = \|B\xi_{95}\|$ of the least *coefficient* witness. The earlier unrestricted result $\eta_{95} = \sqrt{T_0} - \sqrt{\tau} = 1.2885530$ is an image-space comparator; ρ has different units and geometry. The bound $\rho_{95} \ge \eta_{95}/\|B\|_{\mathrm{op}}$ and $q_{95} \ge \eta_{95}$ provide checks.

IV. FINITE-WINDOW RESULTS

The depth response has three projected directions and leaves $T_\perp = 15.3160 > \tau$ even at infinite coefficient radius. The three-column FP proxy leaves $T_\perp = 6.6606$ and crosses at $\rho_{95} = 5.7509$, with $q_{95} = 1.7108$. The six-column joint design spans all six baseline residual directions numerically, leaves a floor below 2×10^{-28}, and crosses at $\rho_{95} = 1.5897$, with $q_{95} = 2.7480$. The large ρ_{sat} values in Table I show that crossing and complete removal are distinct events. The joint image-space minimum $q_{\min} = 1.2885530$ equals the unrestricted radial comparator because the image spans the six-dimensional residual space; its minimum-coefficient witness follows a different route and has the larger q_{95}.

The computation exports the full D and B matrices, singular spectra, whitened residual and nonspanning remainder, threshold coefficient vectors, signed corrections, refitted b, d, and direct GLS residuals. The direct refits reproduce the fixed threshold within 10^{-8}. Synthetic checks recover the wrong block-diagonal baseline, zero gain when $D = X$, nonspanning floors at arbitrarily large budgets, finite equality when the floor equals the threshold, the unrestricted radial formula when $D = L$, and invariance under a simultaneous Cartesian rotation. The computation uses the original source hashes and no source reintegration.

TABLE I. Conditional finite-window designs. All discrepancies use the complete 12D covariance and the unchanged six-degree working threshold. ρ_{95} is a dimensionless joint coefficient budget; q_{95} is its projected whitened image norm. The depth design has no finite threshold crossing.

Design	rank B	$T_\perp$	ρ_{95}	q_{95}	$\rho_{\rm sat}$
Depth slope	3	15.3160	—	—	0.7465
FP composition	3	6.6606	5.7509	1.7108	14.5772
Joint	6	0.0000	1.5897	2.7480	15.4461

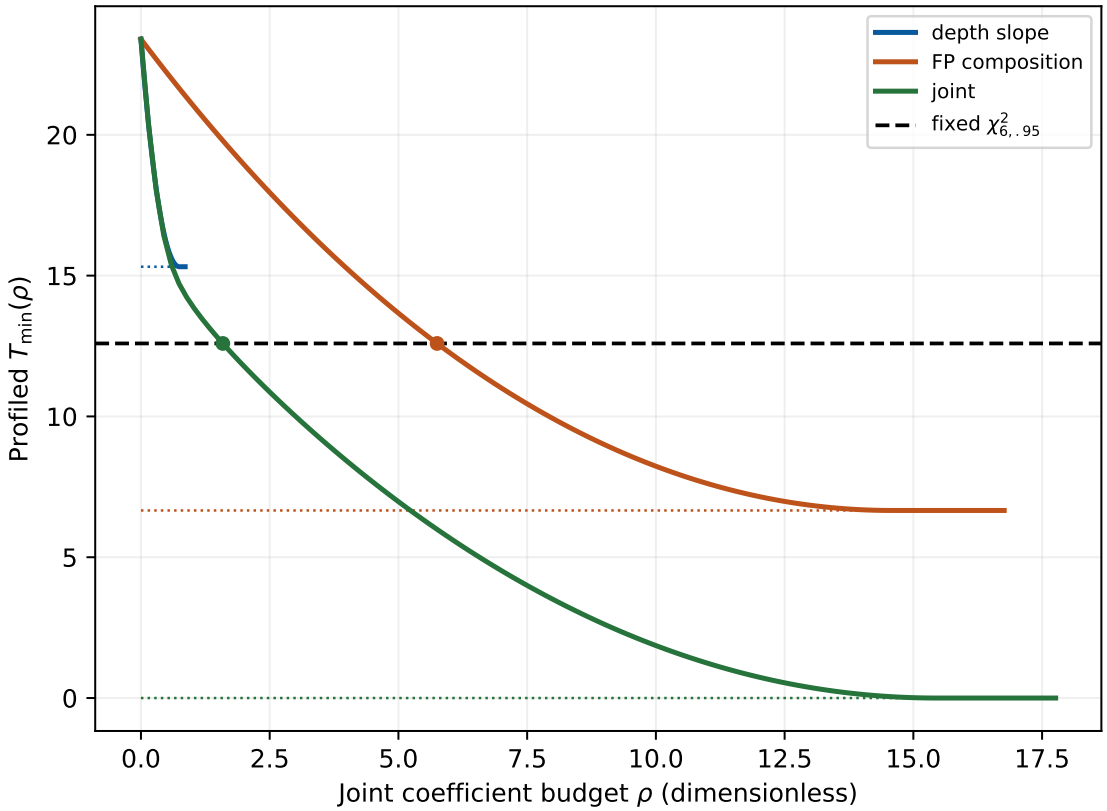


FIG. 1. Profiled discrepancy against a *joint* dimensionless coefficient radius for the three fixed finite-window designs. Dots are attained crossings of the unchanged six-degree reference; dotted horizontal lines are the irreducible floors. The depth-slope curve stays above the reference. Generated from the retained 12D covariance in ICRS J2000.

V. INTERPRETATION AND REQUIRED PHYSICAL INPUT

These results identify sensitivity of a retained finite-window fit to three explicit response spaces. The FP fractions encode sample composition, not a signed calibration Jacobian; family-deletion dipole norms cannot supply such a derivative. For an actual weighted least-squares estimator $\theta = N^{-1}b$, a physical perturbation γ requires $\partial_\gamma\theta = N^{-1}(\partial_\gamma b - (\partial_\gamma N)\theta)$, including any changed distances, design and weights. A source-bound signed window response to named calibration, peculiar-velocity or selection parameters, together with a justified joint bound or law and shared covariance, is needed to interpret η as a physical error. The sky, observer and cosmographic transfer, finite remainder and null coverage are likewise open for stronger claims.

The full physical (x, F, Π, Q, G_F) object remains undefined here; no filling fraction, exceedance probability, score or depth-gap is zero-filled from this 12D statistic. The result supplies a conditional structured-response frontier for a

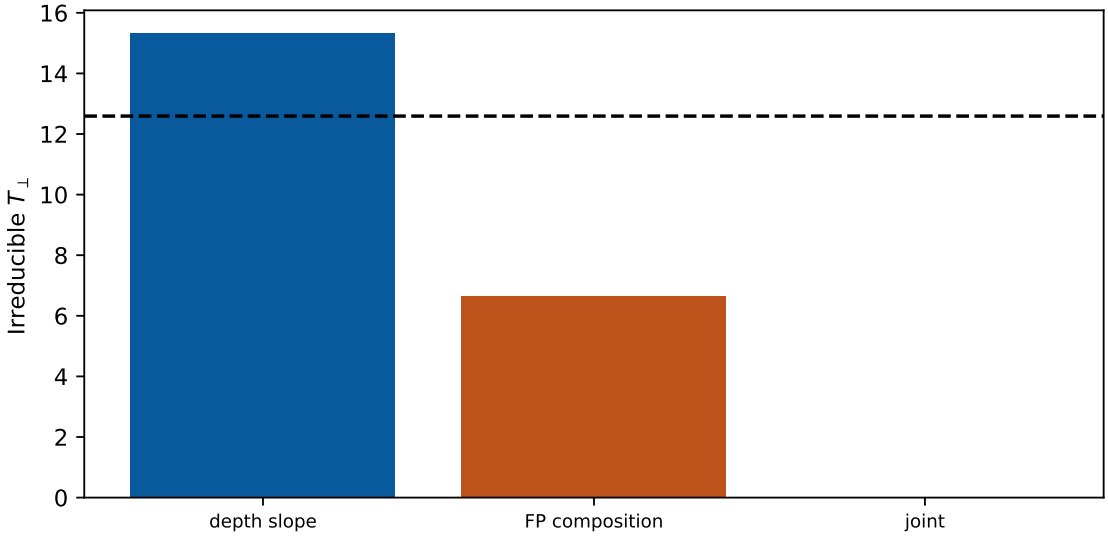


FIG. 2. Exact best attainable discrepancy for each declared projected response subspace. The depth-slope floor exceeds the fixed working threshold. Bars summarize the signed 12-coordinate residual vectors retained in the companion numerical product; they do not encode a physical calibration distribution.

later model-dependent analysis. It does not alter the original 42-node acceptance status or identify a Bianchi family.

[1] Retained H03 source tomography, `evidence/sources/H03_SOURCE_TOMOGRAPHY.json`, SHA-256 `5d31e2ab4fea682c538c31c15181ae7b07e11efba98b36c66aff6feca88f8326` (2026).
[2] Retained physical-results extraction, `evidence/output/physical_results.json`, SHA-256 `25342649e30a8d631a5364002ae39ebffae0a5d92e1e4ad25c9e2f8dfe9362cc` (2026).

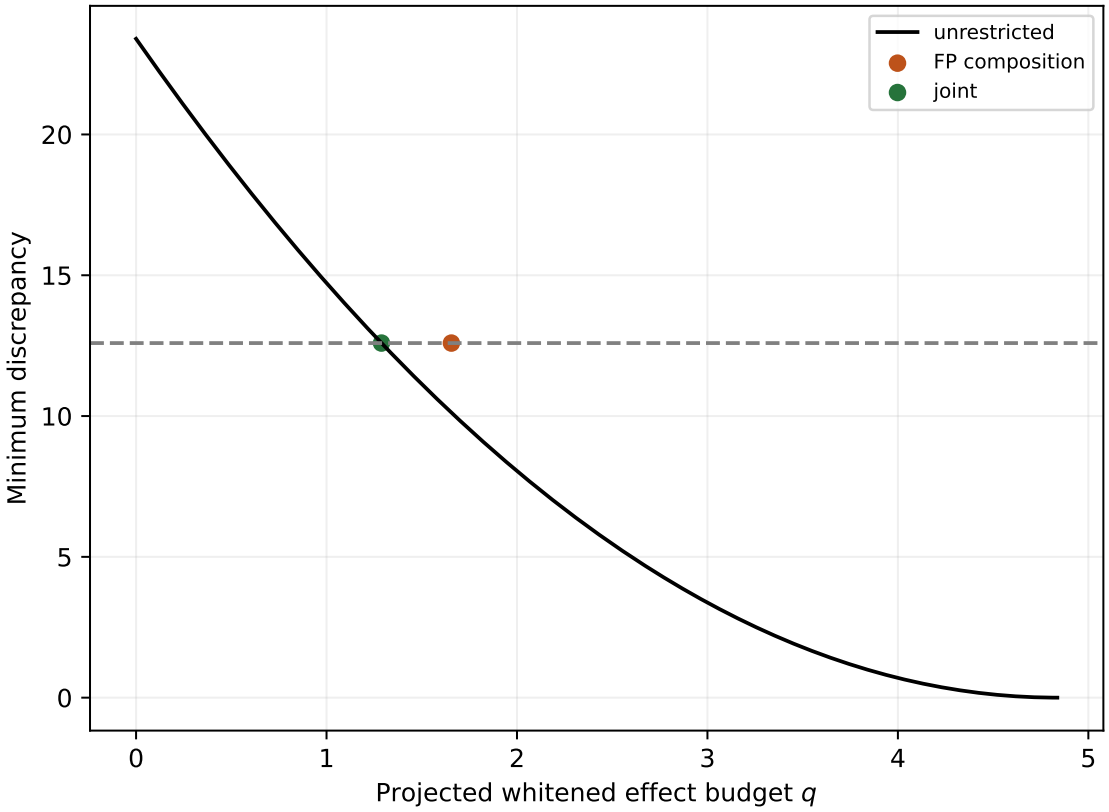


FIG. 3. Unrestricted 12D image-budget frontier $\max(0, \sqrt{T_0} - q)^2$ and the least image radius required by each structure. No depth-slope point exists because its irreducible floor exceeds the threshold. For the FP and joint designs, the plotted minimum-image points are distinct from the coefficient-minimizing image norms q_{95} in Table I.